

Profile

Having 8 years of experience as a Software Engineer in analysis, design, research and development of Windows application, CAD addons. Expertise in Product development and CAD customization and DevOps activities. Analyze application behavior with monitoring tools like Process Monitor, API Monitor, Process explorer and debugging tools like Windows Debugger.

Company: **SECUDE Solutions India Private Limited, Chennai**

Designation: **Senior Software Engineer**

Date: **Sep 2022 – July 2025, July 2019 - Mar 2022**

Product: **HALOCAD (Data Security)**

- HALOCAD provides Rights Management Solutions offering innovative data protection for users of CAD and PLM applications. HALOCAD use DRM solution to protect sensitive data. We use MIP SDK (Siemens NX, AutoCAD) and AD-RMS SDK (Siemens Solid Edge) to exposes the labelling and protection service to third party CAD applications.
- By injecting HALOCAD Hook function into target application, standard OS call which application make are intercepted and using protection information from MIP, we allow or restrict action performed on a file based on the permission of the connected user.
- The following SDK are used for HALOCAD application development: Windows SDK, Microsoft Information Protection, CAD (Application Specific)

Project:

1. **HALOCAD for AutoCAD**
2. **HALOCAD for Siemens NX**
3. **HALOCAD for Siemens Solid Edge**
4. **HALOCAD for EPLAN P8 Electric**

Roles and Responsibilities:

- Responsible for Win32 product development which extend Microsoft Information Protection capabilities to CAD files to protect intellectual property of an organization.
- HALOCAD product is a package of multiple modules which includes following functionalities and my responsibility is to extend these functionalities for the CAD application that I work
 - **File system management (C++ Component)** – This module manages any file management functionalities like CreateFile, ReadFile, WriteFile etc.. and memory management functionalities like CreateFileMapping, MapViewOfFile, UnmapviewOfFile etc.. by hooking standard file management OS calls.

- **UI (VC++ Component)** – This module is provides UI functionality for our HALOCAD product
- **MIP (C++ Component)** - This module contains all Microsoft Purview Information Protection functionalities that encrypt or decrypt file data
- **ADAL (C# Component)** - Microsoft Azure Active Directory Authentication Library (ADAL) module that authenticates client application to Active Directory (Azure or on-premises)
- **CAD (C++ / C# Component)** – This is CAD module which contains CAD related functionalities
- Understanding CAD application flow, extending HALOCAD support for Siemens CAD application like NX, Solid Edge and EPLAN P8 Electric using C++ and C#. We use debugging tools like WinDBG, Visual Studio Debugger and process monitoring application like Process Monitor and API monitor
- Regular customer interaction for to understand client requirements and status update and completion of sprints within the estimated timeline.
- Record and analyze application call stack using monitoring tools like Procmon and set hooking function for OS function calls.
- Responsible for build and deployment activities
 - **Git** – Git tagging, Manage git branches.
 - **Jenkins** – Setup Jenkins job with code coverage
 - **Code quality** – Integration of sonarqube and resolving sonarqube issues.
 - **Product Announcement** – Product announcement to stake holders and QC team.
- Responsible for any bug fixes that are raised by QC team after product announcement.
- Other activities like aligning with Siemens support team (GTAC) for support activities and technical questions by raising GTAC ticket and monitoring GTAC ticket.
- Interact with PLM team (SIEMENS and SAP) to analyze PLM and CAD integrated behavior and further analyze any workflows or requirements

Company: **CLOUDIX Global Solutions Private Limited, Chennai**

Date: **April 2022 to Aug 2022**

Designation: **DevOps Engineer**

Project:

1. **Vera (Health care)**
2. **Mediguru (Health care)**
3. **Palo-Alto (Data Analysis)**

Roles and Responsibilities:

- Responsible for deployment activities for multiple environments (Prod, Test and Dev) for Vera Project with azure services like Azure App service, Azure Kubernetes etc.. (Client: JP.Morgan).
- Responsible for automation activities like key rotation in azure key vault for azure app service and backup previous keys that involves python scripting
- As with agile methodology we have daily stand up call with Vera team and keep them posted with ongoing activities and Completion of sprints within the estimated timeline
- Other activities like ARGO CD implementation for Mediguru environment (deployed in Azure Kubernetes cluster) which helps automated application deployments and application life cycle management. This also helps developers to reduced DevOps team dependencies for household activities like getting logs, monitor status of apps in one dashboard.
- Plan and setup environment for Palo-Alto team to help in development process and other DevOps activities.

Company: P3 DESIGN SOLUTIONS PVT LTD, Chennai

On Site: Mahindra Research Valley, Chennai

Date: June 2017 to July 2019

Designation: Software Engineer

Project:

- 1. NX addon for Configurable design platform (Mahindra and Swaraj)**
- 2. Standard Fastener Selection (Mahindra Knowledge Base Team)**
- 3. Drafting Automation**
- 4. Gear Drawing Automation (Swaraj)**

Roles and Responsibilities:

- Responsible for NX CAD Customization using NXOpen and ufunc with C# and creation of new CAD automation tool as per client requirements.
- Integrate with different teams to understand their pain areas and help them with CAD addons and UI (C# - Winform) enhancements for configurable templates
- As a part of Configurable platform team will discuss CAD design standards and develop checkmate using C#
- Configure Jenkins job for projects and share binaries to deployment team
- Interact with Teamcenter admin team regarding application environment dependencies.
- Involved in 3D modelling, Assemblies, Drafting and Drawing conversion projects using Unigraphics NX10

Skills	Language / Scripting	C++, C# / Shell Scripting, Batch Scripting
	SDK / Framework	CAD, Windows, MIP
	Cloud Management Tool	AWS, Microsoft Azure
	SCM Tool	Git, GitLab
	CI & Deployment Tool	Jenkins, Azure DevOps
	Container Virtualization	Docker, Kubernetes
	Infrastructure Management	Terraform
	Operating System	Windows, Linux
	Ticketing Tool	JIRA
Certification	• Asp.Net developer • HashiCorp Certified: Terraform Associate	
Education	Graduation: Bachelor of Engineering – BE, Mechanical Engineering University: Anna University Institution: SRG Engineering College Secured Marks: 7.7(CGPA) Year of Passing: 2013	

Declaration:

I hereby declare that the above-mentioned information is correct up to my knowledge and I bear the responsibility for the correctness of the Above-mentioned.

Place: Chennai

Date:

Kapil Dev